

Chapter 20

PHY102A

Set #5

Solutions

P1

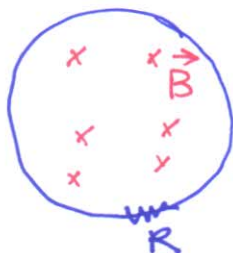
20.1) (a) $P_1 \rightarrow P_2$, Yes, because $\Delta\phi \neq 0$ (b) $P_2 \rightarrow P_3$, No, because $\Delta\phi = 0$ (c) $P_3 \rightarrow P_1$, Yes, because $\Delta\phi \neq 0$

20.2) Direction of $\vec{I}_{ind}$ is counterclockwise because as $P_1 \rightarrow P_2$, $\Delta\phi > 0$, $\vec{I}_{ind}$ will produce $\vec{B}$ opposite to the increase in $\Delta\phi$.

20.5) Given $R = 0.1\text{m}$, $R = 2\Omega$ $B \rightarrow 0 \rightarrow 2\text{T}$
 $t=0$ $t=0.1\text{s}$

$$\mathcal{E} = - \frac{\Delta\phi}{\Delta t} = - \frac{\phi_f - \phi_i}{\Delta t}$$

$$= - \frac{\pi R^2 B_f - \pi R^2 B_i}{t_2 - t_1}$$



$$= - \pi R^2 \frac{(2-0)}{0.1-0} = -\pi(0.1)^2 \frac{2}{0.1} = -0.2\pi$$

$$|\mathcal{E}| = 0.2 \times 3.14 = 6.28 \text{ Volts} \Rightarrow \frac{\mathcal{E}}{R} = \frac{6.28}{2} = 3.14 \text{ A}$$

Question 20.12:

P2

20.12) Given $R = 10 \Omega$, $L = 0.2 \text{ m}$, $f = 100 \text{ Hz}$, $B = 0.5 \text{ T}$

$$(a) \mathcal{E} = - \frac{d\Phi}{dt} = - \frac{d(\vec{B} \cdot \vec{A})}{dt} = - \frac{d(AB \cos \theta)}{dt}$$

$$\theta = \omega t$$

$$\Rightarrow \mathcal{E} = - \frac{d(AB \cos \omega t)}{dt} = \underbrace{\omega AB}_{\text{Amplitude} = \mathcal{E}_m} \sin \omega t$$

$$\mathcal{E}_m = \omega AB = \underbrace{2\pi f}_{\omega} \underbrace{A}_{L^2} B = 2(3.14)10(0.2) \times 0.5$$

$$\mathcal{E}_m = 12.56 \text{ V}$$

$$(b) \mathcal{E}_m = I_{\text{ind}} R \Rightarrow I_{\text{ind.}} = \frac{\mathcal{E}_m}{R} = \frac{12.56}{10} = 1.256 \text{ A}$$

20.17) Given $\mathcal{E}_2 = 50000 \text{ V}$, $\mathcal{E}_1 = 120 \text{ V}$

$$\frac{N_2}{N_1} = \frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{50000}{120} = 417$$

20.16)

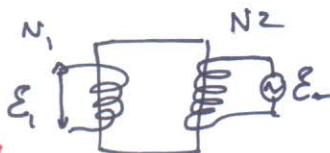
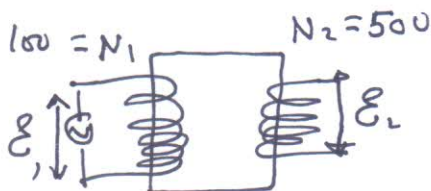
$$(a) \mathcal{E}_1 = 100 \text{ V}$$

$$\mathcal{E}_2 = \frac{N_2}{N_1} \mathcal{E}_1 = \frac{500}{100} \times 120$$

$$\mathcal{E}_2 = 600 \text{ V}$$

$$(b) \mathcal{E}_1 = \frac{N_1}{N_2} \mathcal{E}_2$$

$$\mathcal{E}_1 = \frac{100}{500} \times 120 = 24 \text{ V}$$



20.33)

$$a) \frac{\Delta \Phi}{\Delta t} > 0 \Rightarrow I_{ind} \text{ is Counterclockwise}$$

$$b) \frac{\Delta \Phi}{\Delta t} = 0 \Rightarrow I_{ind} = 0$$

$$c) \frac{\Delta \Phi}{\Delta t} < 0 \Rightarrow I_{ind} \text{ is clockwise}$$

QUESTION 20.6: $R = 10\Omega$, radius = $a = 0.1\text{m}$, $B_i = 3\text{T}$ at $t_i = 0$, $B_f = 3\text{T}$ at $t_f = 0.3\text{s}$. During this time interval the system is rotating from $\theta = 0$ to $\theta = 90^\circ$.

The initial flux at $t_i = 0$ is:

$$\Phi_i = B_i A \cos \theta = B_i \pi a^2 \cos \theta = 3(3.24)(0.1^2) \cos(0) = 0.0942 \text{ Tm}^2$$

$$\Phi_f = B_f A \cos \theta = B_i \pi a^2 \cos \theta = 3(3.24)(0.1^2) \cos(90) = 0 \text{ Tm}^2$$

$$(a) \epsilon = - \frac{\Delta \Phi}{\Delta t} = \frac{\Phi_f - \Phi_i}{t_f - t_i} = - \frac{(0 - 0.0942)}{(0.3 - 0)} = 0.314 \text{ V}$$

$$(b) I_{ind} = \frac{\epsilon}{R} = \frac{0.314}{10} = 0.0314 \text{ A}$$